

Assignment 8 of Machine Learning

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1. Show that the sliced score matching (SSM) loss can also be written as

$$L_{SSM} = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [\|v^T S(x; \theta)\|^2 + 2v^T \nabla_x (v^T S(x; \theta))].$$

We want to show that the sliced score matching (SSM) loss

$$L_{SSM}(\theta) = \mathbb{E}_{x \sim p(x)} \|S(x; \theta)\|^2 + \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [2v^T \nabla_x (v^T S(x; \theta))]$$

can also be written as

$$L_{SSM}(\theta) = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [\|v^T S(x; \theta)\|^2 + 2v^T \nabla_x (v^T S(x; \theta))],$$

where $S(x; \theta)$ is the score function and v is a random vector such that $\mathbb{E}_{v \sim p(v)}[vv^T] = I$.

The two expressions for $L_{SSM}(\theta)$ differ only in the first term. We must show that, under the assumption that the distribution $p(v)$ satisfies $\mathbb{E}_{v \sim p(v)}[vv^T] = I$, the following equality holds:

$$\mathbb{E}_{x \sim p(x)} \|S(x; \theta)\|^2 = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} \|v^T S(x; \theta)\|^2.$$

We focus on the term $\mathbb{E}_{v \sim p(v)} \|v^T S(x; \theta)\|^2$ for a fixed x . Since v and $S(x; \theta)$ are vectors, $v^T S(x; \theta)$ is a scalar. The squared norm is:

$$\|v^T S(x; \theta)\|^2 = S(x; \theta)^T v v^T S(x; \theta).$$

Since $S(x; \theta)$ is independent of v , we can move it outside the expectation operator:

$$\begin{aligned} \mathbb{E}_{v \sim p(v)} \|v^T S(x; \theta)\|^2 &= \mathbb{E}_{v \sim p(v)} [S(x; \theta)^T v v^T S(x; \theta)] \\ &= S(x; \theta)^T (\mathbb{E}_{v \sim p(v)}[v v^T]) S(x; \theta) \\ &= S(x; \theta)^T I S(x; \theta) \\ &= S(x; \theta)^T S(x; \theta) \\ &= \|S(x; \theta)\|^2. \end{aligned}$$

Taking the expectation $\mathbb{E}_{x \sim p(x)}$ on both sides yields:

$$\mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} \|v^T S(x; \theta)\|^2 = \mathbb{E}_{x \sim p(x)} \|S(x; \theta)\|^2.$$

Substituting the new form of the first term back into the original SSM loss:

$$\begin{aligned} L_{SSM}(\theta) &= \mathbb{E}_{x \sim p(x)} \|S(x; \theta)\|^2 + \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [2v^T \nabla_x (v^T S(x; \theta))] \\ &= \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} \|v^T S(x; \theta)\|^2 + \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [2v^T \nabla_x (v^T S(x; \theta))] \\ &= \mathbb{E}_{x \sim p(x)} \mathbb{E}_{v \sim p(v)} [\|v^T S(x; \theta)\|^2 + 2v^T \nabla_x (v^T S(x; \theta))], \end{aligned}$$

this completes the proof. □

2. Briefly explain SDE.

A Stochastic Differential Equation (SDE) is a mathematical tool used to model dynamic systems that evolve under the influence of continuous random noise. It extends the concept of Ordinary Differential Equations (ODEs) by incorporating a random component proportional to the differential of the Wiener process.

The SDE Form and Its Interpretation

The general vector form of an SDE in $\mathbb{R}^d$ is:

$$dx_t = f(x_t, t)dt + G(x_t, t)dW_t, \quad x(0) = x_0,$$

where:

- $x_t \in \mathbb{R}^d$: The stochastic process (the unknown solution) representing the system's state.
- $f(x_t, t) \in \mathbb{R}^d$ (Drift Vector): This is the deterministic component. It defines the instantaneous average rate of change or the expected direction of the process, much like the vector field in an ODE.
- $G(x_t, t) \in \mathbb{R}^{d \times d}$ (Diffusion Matrix): This is the stochastic component. It determines how the random noise affects the system. Specifically, it scales and rotates the driving noise dW_t .
- dW_t : The differential of the Wiener process (or standard Brownian motion) in $\mathbb{R}^d$. This term is the source of the randomness (the "white noise").

The Role of the Wiener Process (W_t)

The Wiener process W_t is crucial. It is characterized by:

- Continuous paths: Its sample paths are continuous functions of time.
- Independent Increments: Changes over disjoint time intervals are independent.
- Gaussian Increments: The increment $\Delta W_t = W_{t+\Delta t} - W_t$ is a Gaussian random variable with zero mean and covariance $\Delta t I$ (i.e., $N(0, \Delta t I)$).

Crucially, W_t is nowhere differentiable. This means the "noise" term dW_t is not a standard differential, which leads to the specialized calculus used for SDEs.

The Itô Integral and the Itô Process

Since dW_t is not a standard differential, the SDE is fundamentally understood through its integral form, defined by the Itô integral:

$$x_t = x_0 + \int_0^t f(x_s, s)ds + \int_0^t G(x_s, s)dW_s.$$

The first integral is a standard Riemann integral. The second integral, $\int G(x_s, s)dW_s$, is the Itô integral. Unlike the standard integral where the function G is evaluated at some intermediate point in the time step, the Itô integral is defined such that the function G is evaluated at the left endpoint of each time interval: $\lim \sum G(x(t_k), t_k)[W(t_{k+1}) - W(t_k)]$. This specific choice is necessary to maintain the martingale property and is the foundation of Itô Calculus. The stochastic process x_t that solves the SDE is called an Itô process.

3. Unanswered Questions

In practical training, we typically use batch processing. When simultaneously sampling multiple x samples and multiple v projection vectors, what is the optimal sampling strategy? For example, for a batch of size B , is it better to sample a unique v_i for each x_i sample, or to share the same v vector across the entire batch? Which strategy provides a better trade-off between computational efficiency and gradient accuracy?